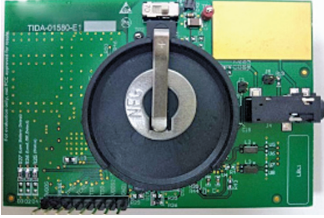
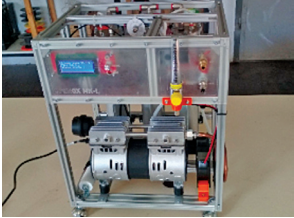


5 Interesting REFERENCE DESIGNS

Title of Reference Design	Reference Design for ECG, SpO2, PTT, and Heart Rate Monitor	Reference Design for Oxygen Concentrator
Image		
Key Attraction	Reference design for a compact wearable device that enables ECG (electrocardiography) and PPG (photoplethysmography). It can also share raw data for SpO2, heart rate, and PPT (partial thromboplastin time). It can work for up to 30 days on a single cell.	Reference design for an open source oxygen concentrator based on molecular sieve and pressure swing adsorption technique. This open source oxygen concentrator can give a maximum oxygen flow of 6 litres per minute (LPM).
Highlights	TIDA-01580-E1 from Texas Instruments (TI) is a reference design of a simple, wearable, multiparameter, patient monitor that is powered by an ARM Cortex M3 microcontroller (MCU). The design employs AFE4900—a single-chip, biosensing front end for measuring synchronised electrocardiography (ECG) and photoplethysmography (PPG). It integrates a CC2640R2F, which adds Bluetooth connectivity to the reference design and also features a 2.4GHz RF transceiver that enables wireless data transfer through BLE 4.2 and 5.0. Apart from measuring ECG and PPG the reference design also provides raw data to calculate SpO2, heart rate, PPT, etc. The reference design features four LEDs and three photodiodes, with digital ambient subtraction to improve the signal-to-noise ratio (SNR). The small form factor and low power requirement make it suitable for implementation in various wearable applications. It can work for up to 30 days with a single cell.	The OpenOx is an open source hardware oxygen concentrator developed primarily by Hacklab, a community-supported hackerspace and improved by various individual contributors. This open source oxygen concentrator design can give a maximum oxygen flow of 6 litres per minute (LPM) with an oxygen concentration level ranging from 85% to 95% while consuming merely 580W of power. The OpenOx concentrator uses a molecular sieve and pressure swing adsorption methodology to produce the oxygen gas output. The oxygen concentrator has an air compressor, two cylinders filled with zeolite granules, a pressure-equalising reservoir, and some valves and tubes. It has a simple design and is optimised for fast production. The module weighs about 20kg and measures 550mm x 400mm x 300mm. All the software of the reference design is released under GNU General Public License V3, and all the hardware are under the CERN Open Hardware License.
Application	The module is suitable for use as a wearable in patient monitoring devices.	This reference design can be used to generate oxygen for patients in hospitals and at homes.
OEM Brand	Texas Instruments (TI)	Hacklab + individual contributors
URL	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-ecg-spo2-ptt-and-heart-rate-monitor	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-an-oxygen-concentrator